

PORTABLE COLORIMETRIC SYSTEM FOR RAPID MILK SPOILAGE DETECTION

1. Introduction

Milk is a highly perishable food product that can easily get spoiled due to microbial growth and biochemical reactions. The traditional methods of detecting milk spoilage, such as microbial culture analysis, titratable acidity test, Methylene Blue Reduction Test (MBRT), require laboratory infrastructure and trained personnel.

There is a need for a rapid, portable, and intelligent milk spoilage detection system that can be easily used in dairy farms and milk collection centres.

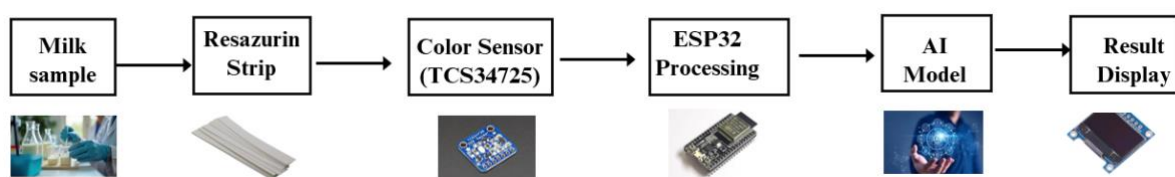
2. Objective

The main objective of this innovation is to design and develop a portable, cost-effective, and AI-enabled milk spoilage detection system that can give real-time quality analysis without depending on any laboratory setup.

3. Proposed Innovation

The proposed innovation combines the following:

- A colorimetric sensing module for detecting chemical reactions that occur in milk spoilage.
- An embedded TinyML machine learning model running on a low-power microcontroller.
- A portable hardware module that can perform real-time analysis.
- The colour change that occurs as a result of reactions in milk spoilage is detected and analysed by the embedded module. The TinyML model detects the patterns of colour change and makes predictions about the level of spoilage without the need for cloud connectivity.



4. Innovative Features

- Integration of colorimetric sensing with embedded TinyML inference.
- Real-time detection of spoilage based on low-power microcontrollers.
- Portable and field-ready system.
- No need for laboratory setup.
- Energy-efficient and cost-effective system.
- Applicable in rural and resource-poor dairy settings.

5. Advantages Over Existing Technologies

- Faster detection than laboratory microbial analysis.
- Intelligent decision-making capability with on-device AI.
- Lower operational cost.
- Scalable for dairy supply chain applications.
- Prevents economic loss from spoiled milk distribution.

6. Industrial & Commercial Potential

The proposed innovation can be applied in dairy farms, milk collection points, food processing companies, cold storage facilities, and dairy supply chain checkpoints. There is a strong need for food safety and rapid quality analysis; the proposed innovation has great commercial potential in the dairy industry.

7. Current Status

Prototype development and TinyML model integration are in progress for testing and performance optimisation.